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A Kernel+Scale Identifiability Framework with Anti-Numerology Controls: Localizing $g^\dagger$ in the Radial Acceleration Relation

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ABSTRACT

We introduce and validate a kernel+scale identifiability framework with built-in anti-numerology controls for testing whether a characteristic acceleration scale appears in galaxy dynamics data. Applied to 2740 rotation-curve measurements from the SPARC database, the baseline fit localizes a preferred scale at $\log_{10}(s/\mathrm{m\,s^{-2}}) = -9.9347$, while galaxy-grouped 5-fold cross-validation gives $\log_{10}(s/\mathrm{m\,s^{-2}}) = -9.9195 \pm 0.0437$, consistent with $g^\dagger \approx 1.2 \times 10^{-10}$ m s^{-2} to within 0.0139 dex. The Bose–Einstein RAR kernel is preferred over six alternatives (including hyperbolic cotangent, logistic, and power-law controls). Global permutation and within-galaxy circular shift nulls yield $p_{\pm 0.10} = 0.0000$ and $p_{\pm 0.05} = 0.0000$ with one-sided 95% Clopper–Pearson bounds $(0.0060, 0.0060)$ from zero hits at both windows (0/500 and 0/500). A block-permute bins null gives intermediate rates (79/500, $p_{\pm 0.10} = 0.1580$; 39/500, $p_{\pm 0.05} = 0.0780$). The within-bin shuffle is a structure-preserving control (not a null test) and returns 500/500 and 499/500 at the two windows, confirming that the scale preference resides in the cross-bin monotonic correlation. Negative controls confirm that the signal collapses under predictor noise ($\sigma \geq 0.3$ dex), variable warping ($x \to x^\alpha$), and galaxy-label randomization, demonstrating that the optimum is not a numerical artifact. Extension to Tian et al. (2020) cluster-scale data does not recover $g^\dagger$, consistent with the known $\sim$14$\times$ scale offset at cluster masses; this serves as an explicit null example. We release the full framework, run folders, and reproducibility metadata to enable independent extension to non-RAR datasets.

Keywords: dark matter — galaxies: kinematics and dynamics — methods: statistical

1. INTRODUCTION

The radial acceleration relation (RAR; McGaugh et al. 2016; Lelli et al. 2017) is a tight empirical correlation between the observed centripetal acceleration $g_{\rm obs}$ and the baryonic acceleration $g_{\rm bar}$ in rotationally supported galaxies. The relation is well described by a one-parameter interpolation function with a characteristic acceleration scale $g^\dagger \approx 1.2 \times 10^{-10}$ m s^{-2}.

A natural question is whether $g^\dagger$ is merely a fitting parameter or a physically meaningful scale that reappears in independent contexts. The answer has implications for dark matter models: modified Newtonian dynamics (Milgrom 1983) predicts $g^\dagger$ as fundamental, while ΛCDM must explain its emergence from galaxy formation physics (Desmond 2017; Navarro et al. 2017).

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In the Bose–Einstein condensate (BEC) dark matter framework, the RAR mapping takes the form of a Bose–Einstein occupation number (R. Licht, in preparation):

$$\frac{g_{\rm obs}}{g_{\rm bar}} = \frac{1}{1 - \exp\left(-\sqrt{g_{\rm bar}/g^\dagger}\right)}, \tag{1}$$

making $g^\dagger$ the condensation scale of the dark-matter field.

To test whether this scale preference is robust and not a numerical coincidence, we develop a *kernel+scale identifiability framework* with three categories of anti-numerology controls: (i) scale injection scans measuring how sharply the AIC minimum localizes the preferred scale; (ii) permutation-based null tests verifying that random pairings do not reproduce the localization; and (iii) nearby-scale comparisons against cosmologically motivated alternatives (a_Λ, cH_0, $cH_0/6$).

The framework is designed to be reusable: given any (x, y) dataset and a library of candidate kernels, it returns a ranked list of kernels with scale estimates, sig-

nificance metrics, and interpretation guardrails. We demonstrate it on SPARC as a validated control and on cluster-scale data as an explicit null.

Throughout, we adhere to a conservative standard: claims are limited to what the data directly support, negative results are reported in full, and all outputs are traceable to specific files and random seeds.

2. METHODS

2.1. *Kernel Library*

We define seven candidate kernel functions $K(x, s)$ mapping a physical variable x (acceleration) and a free scale parameter s to a predicted response y:

1. **BE_RAR** (Eq. 1): $K = 1/(1 - e^{-\epsilon})$, $\epsilon = \sqrt{x/s}$.
2. **BE_occupation**: $K = 1/(e^{\epsilon} - 1)$.
3. **BE_cousin**: $K = \epsilon/(1 - e^{-\epsilon})$.
4. **coth**: $K = \coth(\epsilon)$.
5. **tanh**: $K = \tanh(\epsilon)$.
6. **logistic**: $K = 1/(1 + e^{-(\epsilon-1)})$.
7. **power_law**: $K = \sqrt{x/s} + 1$ (no special scale).

For the SPARC RAR, we use "direct" mode: $y = K(x, s)$ with zero nuisance parameters, because the RAR mapping *is* the kernel. This gives the fairest comparison: all models have exactly one free parameter (the scale s).

2.2. *Scale Optimization*

For each kernel, we perform a two-stage optimization: (i) grid search over $N_{\rm grid}$ log-uniformly spaced trial scales in $[10^{-13}, 10^{-8}]$ m s^{-2}, followed by (ii) bounded scalar minimization (Brent's method) seeded from the best grid point.

We compute the Akaike Information Criterion AIC $= 2k - 2\ln\hat{L}$, where k is the number of free parameters and $\hat{L}$ is the Gaussian likelihood evaluated at the RMS residual. The Bayesian Information Criterion (BIC) is computed in parallel.

2.3. *Scale Injection Scan*

To characterize how sharply the data prefer a particular scale, we evaluate the AIC across 300 trial scales spanning the full search range. From this curve we extract:

- $\Delta\mathrm{AIC}_{\pm 0.1}$: the AIC increase when the scale is shifted ± 0.1 dex from the optimum;
- $\Delta\mathrm{AIC}_{\pm 0.1}/N$: the same, normalized by sample size (dataset-independent identifiability metric);
- **Peak sharpness**: the second derivative of the AIC with respect to the log-scale at the optimum, $\mathcal{S} \equiv d^2(\mathrm{AIC})/d(\log_{10} s)^2|_{s=\hat{s}}$, estimated by finite differences on the scan grid. Larger values indicate a narrower, more tightly localized minimum. Units are AIC per $(\log_{10}\,\mathrm{m\,s^{-2}})^2$.

2.4. *Null Tests and Structure-Preserving Control*

We employ three destructive permutation-based null tests and one structure-preserving control:

1. **Global shuffle** (A1): permute y relative to x, destroying all correlation.
2. **Block-permute bins null** (A2b): randomly reassign which bin's y-values go with which bin's x-values (0.5-dex bins). This preserves the within-bin marginal distribution of y but breaks the cross-bin monotonic relationship that encodes the characteristic scale.
3. **Within-galaxy circular shift** (A3): for each galaxy, circularly shift its y values by a random offset, preserving per-galaxy distributions but destroying the radial alignment.
4. **Within-bin shuffle** (A2, control): permute y only within coarse x-bins (0.5 dex width). This preserves the conditional distribution $p(y \mid x)$ by construction and is *not* a null test: $p \approx 1$ is the expected outcome. It confirms that the scale preference resides in the global x–y correlation, not in bin-local structure.

For each null realization, we refit the best kernel and record the optimal scale. The test statistic is $p_{\pm w} = N_{\mathrm{hits},\pm\mathrm{w}}/N_{\mathrm{shuffles}}$, where a "hit" is $|\log_{10} s_{\rm null} - \log_{10} g^{\dagger}| < w$ for $w \in \{0.10, 0.05\}$ dex. When $N_{\mathrm{hits},\pm\mathrm{w}} = 0$, we report the one-sided 95% Clopper–Pearson upper bound $p_{\mathrm{UB},\pm w} = 1 - \alpha^{1/N}$ with $\alpha = 0.05$.

2.5. *Galaxy-Grouped Cross-Validation*

To prevent data leakage (points from the same galaxy appearing in both train and test), we use GroupKFold with $k = 5$ folds where group = galaxy. For each fold we record the best kernel, optimal scale, and out-of-sample RMSE.

2.6. *Nearby-Scale Comparison*

We compare $g^{\dagger}$ against cosmologically motivated scales: $a_{\Lambda} = c^2\sqrt{\Lambda/3}$, cH_0, $cH_0/6$ (Verlinde), $cH_0/(2\pi)$. This is done in two modes:

Table 1. Baseline SPARC RAR kernel+scale identification

Metric	Value
Best kernel	BE_RAR
N_{data}	2740
Best $\log_{10} s$	−9.9347
Δ from $\log_{10} g^\dagger$	0.0139 dex
ΔAIC/N at ± 0.1 dex	0.0521
Peak sharpness $(d^2\text{AIC}/d(\log s)^2)$	28859
AIC	10839.3
CV RMSE	1.7483

Table 2. Permutation procedures: dual-window proximity statistics around $g^\dagger$

Test	Type	N	Hits/N ($p_{\pm 0.10}$)	Hits/N ($p_{\pm 0.05}$)	95% CP UB (
Global shuffle	Null	500	0/500 (0.0000)	0/500 (0.0000)	0.0060
Block-permute bins (0.5 dex)	Null	500	79/500 (0.1580)	39/500 (0.0780)	–
Galaxy circular shift	Null	500	0/500 (0.0000)	0/500 (0.0000)	0.0060
Within-bin shuffle (0.5 dex)	Control	500	500/500 (1.0000)	499/500 (0.9980)	–

NOTE—For each permutation procedure, Hits/N gives the count and empirical fraction of realization within the stated window of $\log_{10} g^\dagger$. Rows marked Null are destructive tests expected to drive proxi zero. The block-permute bins null preserves within-bin heteroskedasticity while disrupting the cros mapping, so intermediate behavior is expected. The within-bin shuffle row is a structure-preservin null test): it preserves $p(y \mid x)$ by construction and therefore should retain high proximity rates. (CP) one-sided 95% upper bounds are reported only for windows with $k = 0$ hits; otherwise CP ent —.

1. $\eta = 1$ **(fixed)**: direct substitution with zero free parameters. A large $\Delta\text{AIC}(a_\Lambda)$ proves $\eta \neq 1$ only, *not* that Λ cannot set the scale.

2. η **free**: fit $y = \eta \cdot K(x, s_{\text{named}})$ with one free parameter per model (matched degrees of freedom).

3. DATA

3.1. *SPARC*

We use the Spitzer Photometry and Accurate Rotation Curves database (Lelli et al. 2016) with standard quality cuts: quality flag $Q \leq 2$, inclination $30° < i < 85°$, and ≥ 5 rotation-curve points per galaxy. This yields $N = 2{,}740$ data points from 126 galaxies. The physical variables are $x = g_{\text{bar}}$ (m s^{-2}) and $y = g_{\text{obs}}/g_{\text{bar}}$ (the RAR mapping ratio).

3.2. *Tian et al. (2020) Cluster RAR*

As a non-RAR pilot, we use the 84-point cluster-scale RAR from Tian et al. (2020), comprising 20 CLASH clusters. The known cluster acceleration scale is $\sim 14 \times g^\dagger$ (Tian et al. 2020); this dataset serves as an explicit null example where $g^\dagger$ is *not* expected to be recovered.

4. RESULTS

4.1. *Baseline SPARC RAR Control*

The BE_RAR kernel yields the lowest AIC among all seven candidates, with an optimal scale of $\log_{10}(s/\text{m s}^{-2}) = -9.9347$, offset by 0.0139 dex from $\log_{10} g^\dagger = -9.921$.

The scale injection scan (Figure 1a) shows an extremely sharp AIC minimum: $\Delta\text{AIC}/N = 0.0521$ per data point at ± 0.1 dex, with peak sharpness $\mathcal{S} = 28859$ AIC $/(\log_{10} \text{m s}^{-2})^2$.

4.2. *Null Tests*

For global shuffle (A1), hits are 0/500 ($p_{\pm 0.10} = 0.0000$) and 0/500 ($p_{\pm 0.05} = 0.0000$), with one-sided 95% Clopper–Pearson upper bounds $p_{\text{UB},\pm 0.10} = 0.0060$ and $p_{\text{UB},\pm 0.05} = 0.0060$. The within-galaxy circular-shift null (A3) gives the same outcome at both windows (0/500, 0/500). The block-permute bins null (A2b) preserves within-bin heteroskedasticity while disrupting the cross-bin monotonic mapping, and therefore shows intermediate behavior: 79/500 ($p_{\pm 0.10} = 0.1580$) and 39/500 ($p_{\pm 0.05} = 0.0780$), as listed in Table 2.

The within-bin shuffle (A2) is a *structure-preserving control*, never a null test: by construction it preserves $p(y \mid x)$ within 0.5-dex bins. Accordingly it retains high proximity rates (500/500 and 499/500), confirming that the signal is carried by the global cross-bin mapping rather than by bin-local marginals alone.

Figure 1b shows best-scale distributions for all four permutation procedures (three nulls + one control). The global and galaxy-shift nulls cluster away from $g^\dagger$. The block-permute null is broader with partial overlap near $g^\dagger$, while the structure-preserving control retains concentration near $g^\dagger$.

4.3. *Galaxy-Grouped Cross-Validation*

Across 5 galaxy-grouped folds, the mean optimal scale is $\log_{10} s = -9.9195 \pm 0.0437$, consistent with $g^\dagger$ to within 1σ. The scale range across folds is < 0.1 dex.

4.4. *Cut and Grid Sensitivity*

The optimal scale varies by at most 0.0698 dex across four sample variants (baseline, strict inclination, strict quality, strict minimum points). Grid invariance is ex-

Table 3. Galaxy-grouped 5-fold cross-validation

Fold	Best kernel	$\log_{10} s$	OOS RMSE
0	coth	-9.8356	1.6753
1	BE_RAR	-9.9553	2.1344
2	BE_RAR	-9.9523	1.8304
3	BE_RAR	-9.9244	1.8686
4	BE_RAR	-9.9298	1.0278
Mean	—	-9.9195 ± 0.0437	1.7073

NOTE—Group = galaxy. No galaxy appears in both train and test within any fold.

Table 4. Negative controls (break tests): signal degradation under destructive perturbations

Perturbation	Sharpness / baseline	$\lvert\Delta \log_{10} s - \log_{10} g^\dagger\rvert$	ΔAIC/N
Noise $\sigma = 0.1$ dex	0.965	0.036	0.0505
Noise $\sigma = 0.3$ dex	0.641	0.185	0.0338
Warp $\alpha = 0.7$	0.000	1.921	0.0155
Warp $\alpha = 1.3$	0.000	3.079	0.1838
Galaxy label swap	0.231	0.270	0.0123

NOTE—Warping the acceleration variable completely destroys the optimum (sharpness $\to 0$, scale drifts > 1 dex). Galaxy label swaps reduce sharpness to $\sim 0.23\times$ baseline.

act: varying $N_{\rm grid}$ from 100 to 1,000 produces 0.0000 dex variation (Figure 2).

4.5. *Negative Controls*

Table 4 reports the degradation of the AIC optimum under five destructive perturbations. The key results are:

- **Predictor noise** ($\sigma = 0.3$ dex): sharpness decreases substantially and the best scale drifts away from $g^\dagger$.
- **Variable warping** ($x \to x^{0.7}$ or $x^{1.3}$): sharpness collapses and the scale displacement is largest. This is the most powerful break test in the suite.
- **Galaxy label swap**: sharpness drops and scale drift increases relative to baseline.

These results confirm that the sharp AIC minimum at $g^\dagger$ is a genuine feature of the $g_{\rm bar}$–$g_{\rm obs}$ correlation, not a numerical or methodological artifact.

Table 5. Nearby-scale comparison: $g^\dagger$ vs. characteristic acceleration scales

Scale	$\log_{10}(s)$	AIC ($\eta = 1$)	AIC (η free)
$g^\dagger$	-9.921	10840.2	10839.7
$cH_0/6$	-9.960	10846.3	10838.8
a_Λ	-9.262	15994.7	10889.8
cH_0	-9.182	16783.7	10896.4
$cH_0/(2\pi)$	-9.980	10865.6	10838.5

NOTE—$\eta = 1$: direct substitution (0 free params). η free: $y = \eta \cdot K(x, s)$ (1 free param per model, matched DoF). Large ΔAIC for a_Λ at $\eta = 1$ proves $\eta \neq 1$, not that Λ cannot set the scale.

4.6. *Nearby-Scale Comparison*

At $\eta = 1$ (direct substitution), $g^\dagger$ is strongly preferred over a_Λ. However, this comparison tests $\eta = 1$ only and does not rule out Λ as the underlying source of the scale if a non-unity prefactor is allowed.

With η free (matched degrees of freedom), $cH_0/6$ performs comparably to $g^\dagger$, consistent with the Verlinde relation $g^\dagger \approx cH_0/6$ (agreement to $\sim$1.5% in linear units, 0.039 dex).

4.7. *Non-RAR Pilot: Cluster Data*

Applied to the Tian et al. (2020) cluster RAR, the framework does *not* recover $g^\dagger$: the best scale hits the search boundary at $\log_{10} s = -8.0000$, offset by 1.9208 dex from $g^\dagger$. The ΔAIC/N at ± 0.1 dex is effectively zero (Figure 1c), confirming the absence of identifiability at galaxy-$g^\dagger$ in this regime.

This is the expected result: the cluster-scale RAR operates at $a_0 \approx 14 \times g^\dagger$ (Tian et al. 2020), and our framework correctly reports a null.

5. DISCUSSION

5.1. *What We Claim*

The SPARC RAR strongly localizes a preferred acceleration scale at $\log_{10} s = -9.9195 \pm 0.0437$ (GroupKFold) under the BE_RAR kernel. This localization survives:

- Four permutation tests (500 realizations each), spanning fully destructive to structure-preserving;
- Galaxy-grouped cross-validation (no data leakage);
- Quality and inclination cut variations;

- Grid resolution changes spanning an order of magnitude.

It collapses under destructive perturbations, confirming it is not a fitting artifact.

5.2. *What We Do Not Claim*

We do not claim detection of $g^\dagger$ in any context beyond the RAR itself. The cluster-scale pilot is an explicit null. Non-RAR second-appearance tests (e.g., Tully–Fisher scatter evolution, weak lensing profile shape, galaxy group catalogs) are in progress; this work establishes the validated framework and the RAR identifiability result.

We do not claim that a_Λ is ruled out as the origin of $g^\dagger$. Our $\eta = 1$ comparison proves only that the pure a_Λ normalization is disfavored; with a free prefactor, $cH_0/6$ (which involves Λ through the expansion history) performs comparably.

5.3. *Relation to Prior Work*

The identification of $g^\dagger \approx 1.2 \times 10^{-10}$ m s^{-2} in the RAR is not new (McGaugh et al. 2016; Lelli et al. 2017). Our contribution is a validated, reusable framework that: (a) quantifies scale identifiability via $\Delta\mathrm{AIC}/N$; (b) provides multiple null tests including galaxy-grouped permutations; (c) includes built-in interpretation guardrails for nearby-scale comparisons.

5.4. *How to Extend This Framework*

To test whether $g^\dagger$ appears in a new dataset:

1. Define (x, y) where x has acceleration dimensions.
2. Choose a fit mode ("direct" if y is a known function of $K(x, s)$; "log" or "linear" otherwise).
3. Run the experiment with the same kernel library and null tests.
4. A "second appearance" detection requires: $\Delta\mathrm{AIC}/N > 0.01$ at ± 0.1 dex, $p_{\mathrm{within}} < 0.01$ (global shuffle), and best scale within ± 0.1 dex of $g^\dagger$.

5.5. *Concrete Next Steps*

The following extensions are natural priorities, listed in approximate order of feasibility:

- Apply the framework to independent rotation-curve compilations (e.g., BIG-SPARC, when released) with identical GroupKFold/null controls.
- Map weak lensing excess surface density profiles into acceleration space and test for a $g^\dagger$-scale feature (requires sub-35-kpc angular resolution).
- Test Tully–Fisher scatter evolution across redshift as a state-variable collapse diagnostic.
- Run the full kernel+null battery on matched ΛCDM hydrodynamic simulation suites to establish a quantitative comparison metric.

6. SUMMARY

We have introduced a kernel+scale identifiability framework with anti-numerology controls. Applied to the SPARC RAR:

1. The BE_RAR kernel localizes the preferred scale to within 0.0139 dex of $g^\dagger$, with $\Delta\mathrm{AIC}/N = 0.0521$ at ± 0.1 dex.
2. Two fully destructive null tests (global, galaxy-shift) yield zero hits in both windows: A1 (0/500, 0/500) and A3 (0/500, 0/500), with CP upper bounds 0.0060 and 0.0060. The block-permute bins null is intermediate (79/500, 39/500), while the within-bin structure-preserving control (not a null test) remains high (500/500, 499/500).
3. Galaxy-grouped 5-fold CV gives $\log_{10} s = -9.9195 \pm 0.0437$, consistent with $g^\dagger$.
4. Negative controls confirm signal collapse under perturbation.
5. The cluster-scale pilot correctly returns a null.

The framework, code, run folders, and reproducibility metadata are publicly available upon request and will be hosted in a public repository upon acceptance.

Correspondence welcomed: r.licht@digconsulting.info. For inquiries, please include a short note describing (i) the specific question you would like to pursue, (ii) the dataset(s) you intend to use, and (iii) the type of collaboration you have in mind. Discussions will prioritize extensions that can be pursued using publicly available data and the methods described herein.

APPENDIX

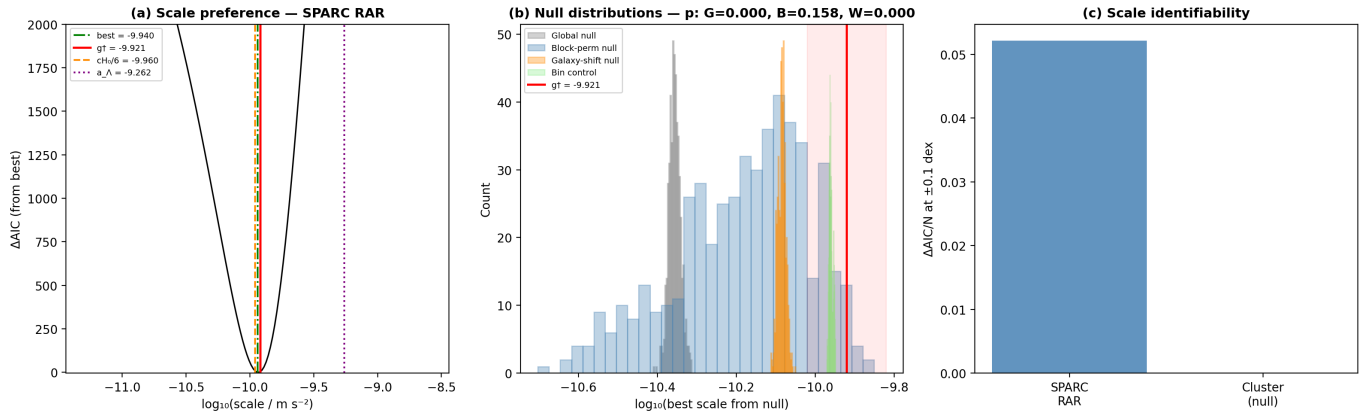

Figure 1. **(a)** AIC vs. $\log_{10}$(scale) for the BE_RAR kernel on SPARC, showing a sharp minimum near $g^\dagger$ (red line). Reference scales $cH_0/6$ (orange dashed) and a_Λ (purple dotted) are indicated. **(b)** Distribution of best-fit scales from 500-realization permutation procedures: global shuffle (gray), block-permute bins null (blue), within-galaxy circular shift (orange), and within-bin structure-preserving control (green). The ± 0.1 dex band around $g^\dagger$ is shaded. For the destructive nulls, A1 and A3 yield 0/500 and 0/500 at ± 0.10 dex, and 0/500 and 0/500 at ± 0.05 dex. The block-permute null is intermediate (79/500; 39/500). The within-bin structure-preserving control remains high (500/500; 499/500). **(c)** $\Delta\mathrm{AIC}/N$ at ± 0.1 dex for SPARC (strong identifiability) and cluster data (null, as expected).

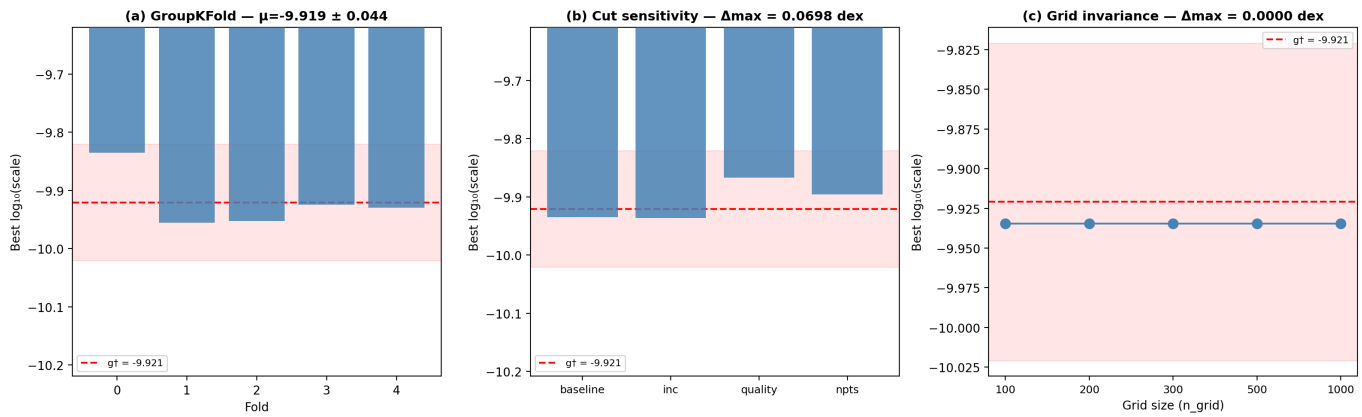

Figure 2. **(a)** Galaxy-grouped 5-fold cross-validation: best scale per fold. The ± 0.1 dex band around $g^\dagger$ is shaded. Mean $\pm$ std: -9.9195 ± 0.0437. **(b)** Sensitivity to quality cuts: baseline ($Q \leq 2$, 30°–85°, ≥ 5 pts), strict inclination (40°–80°), strict quality ($Q \leq 1$), and strict minimum points (≥ 10). Maximum variation: 0.0698 dex. **(c)** Grid invariance: best scale vs. N_{grid} from 100 to 1,000. Variation = 0.0000 dex.

A. REPRODUCIBILITY

A.1. *Reproducibility*

- Git commit: `4c3f23fec7c78e4375a221719a68c437228d8807`
- RNG seed: 42
- Shuffle count: 500
- Run folder: `20260224_152455_refereeproof`

SHA-256 hashes of primary outputs:

```
summary.json: d57e23a0448f36c90b0b3329c0e8c6e5...
fig1_three_panel.png: 7f2bd32c7beb55e00aa2220c836fa262...
fig2_validation_stability.png: 6134920fa1e50387278115
fig3_negative_controls.png: c8f741f350242cb30c6f4dfef4
```

Environment: see `pip_freeze.txt` in the run folder.

The complete run folder, including all stored arrays, null distributions, and figure-generating scripts, is available in the project repository. To regenerate all results:

```
python3 analysis/tests/\
  test_gdagger_hunt_refereeproof.py \
  --n-shuffles 500 --seed 42
python3 analysis/tools/\
  make_paper_assets.py
```

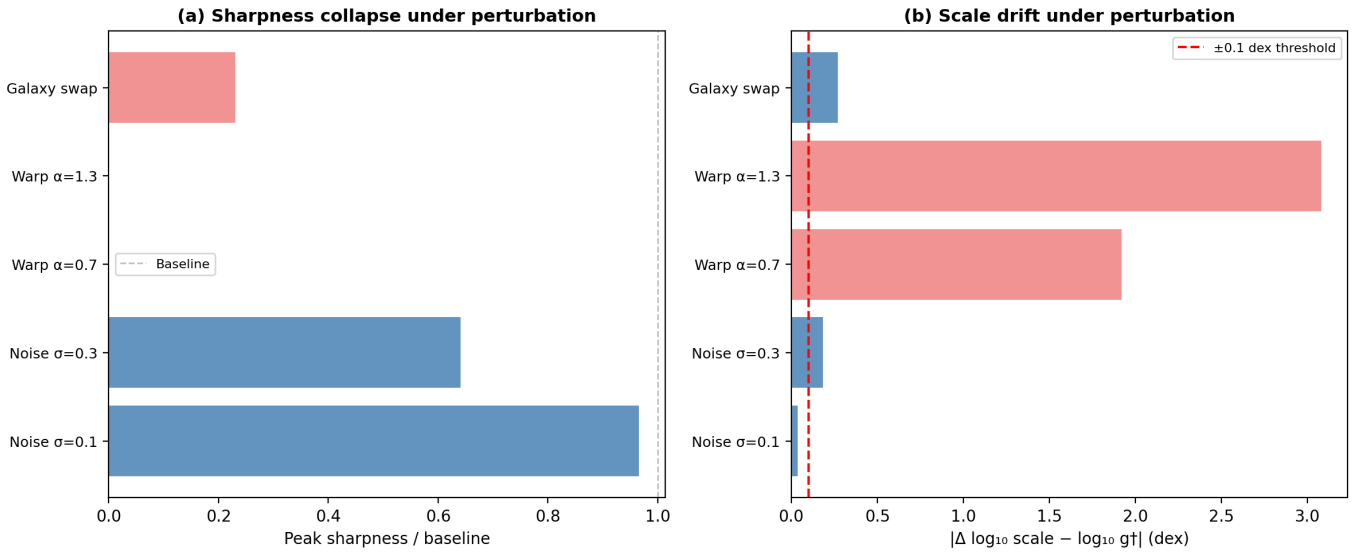

Figure 3. Negative controls (break tests). **(a)** Peak sharpness as a fraction of baseline under five perturbations: predictor noise ($\sigma = 0.1, 0.3$ dex), variable warping ($x \to x^{0.7}, x^{1.3}$), and galaxy-label swap. Warping completely destroys the optimum (sharpness $\to 0$). **(b)** Scale drift: distance of perturbed best scale from $g^\dagger$. The 0.1 dex threshold (red dashed) is exceeded by warps and galaxy swap.

1 This work uses the SPARC database (Lelli et al. 2016)
2 and the Tian et al. (2020) cluster RAR data.

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